

WASP-12b

R-2457

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_140402 · created 2026-08-23T00:30:15+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456749.662911 +/- 3.8e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.017
Transit depth (Rp/Rs)^2	1.32 +/- 0.4 [%]
Orbital Inclination (inc)	85.11 +/- 1.92
Airmass coefficient 1 (a1)	1.293 +/- 0.01
Airmass coefficient 2 (a2)	-0.1265 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	None
Optimal Aperture	4.49
Optimal Annulus	5.85
Transit Duration (day)	0.0881 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.017 vs 0.1178 (deviation 0.15 σ)
Duration fit vs published	0.0881 d vs 0.125 d (deviation 19.7 σ)
Mid-time O-C	0.1 min
Depth SNR	6.1
Residual scatter	0.8 %

Light curve

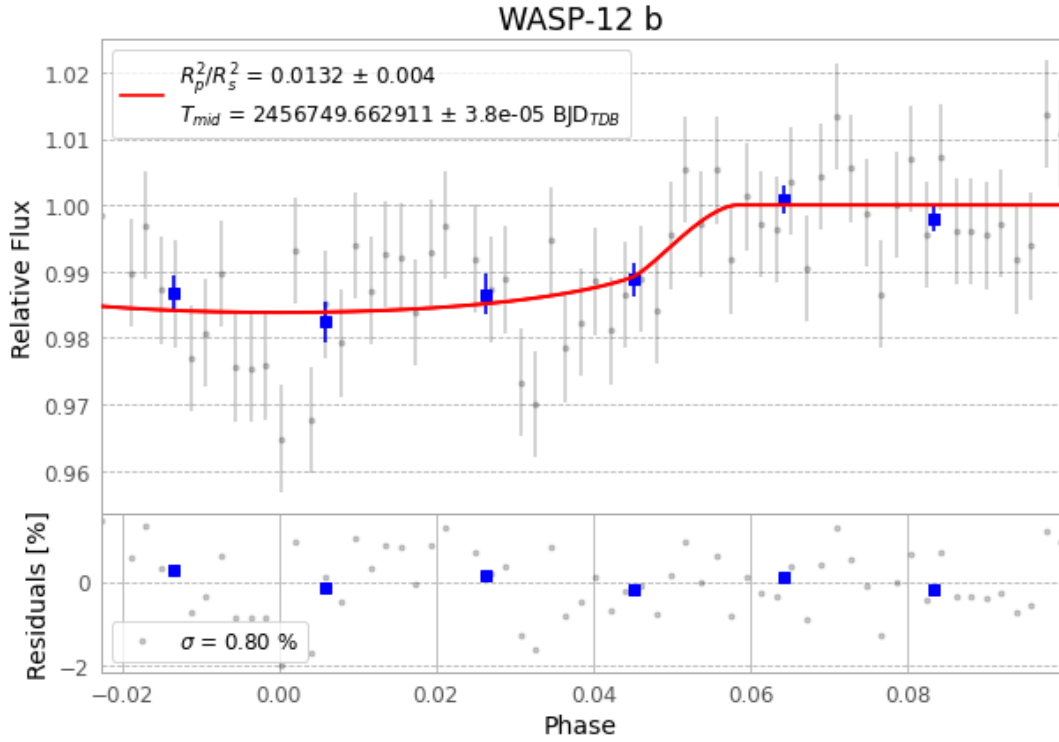


Figure 1 — FinalLightCurve_WASP-12 b_2014-04-01.png (SHA-256 66f891c9c6415580...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 275], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 23d7894863b72935...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), WASP-12140402031212.FITS → WASP-12140402063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-140402.log (3fa3208b3b4a...), FinalLightCurve_WASP-12 b_2014-04-01.png (66f891c9c641...), FinalLightCurve_WASP-12 b_2014-04-01.csv (9133e4f47ed5...)

Compute resources

Wall time 87.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:30Z.